SCHOOL MANAGEMENT API
PHASE 1 — MISSION SPECIFICATIONS

Scenario

You are a software developer building a School Management System Web API. Your mission is to design the initial domain model for the system using ASP.NET Core Web API and Entity Framework Core. In this phase, you will create the core entities, define their relationships, apply validation rules, and configure the EF Core model.

System Models

Department

Property	Data Type	Validation Rules
Id	int	Primary key
Name	string	Required, maximum length 100
Description	string	Optional (nullable), maximum length 500

Teacher

Property	Data Type	Validation Rules
Id	int	Primary key
FirstName	string	Required, maximum length 50
LastName	string	Required, maximum length 50
Email	string	Required, valid email address, maximum length 150
PhoneNumber	string	Optional (nullable), maximum length 20, valid phone number if provided
Salary	decimal	Required, must be greater than or equal to 0

Subject

Property	Data Type	Validation Rules
Id	int	Primary key
Name	string	Required, maximum length 100
Description	string	Optional (nullable), maximum length 500
MaxGrade	int	Required, value between 1 and 100

ClassRoom

Property	Data Type	Validation Rules
Id	int	Primary key
Name	string	Required, maximum length 50, cannot be empty
GradeLevel	int	Required, value between 1 and 12
Capacity	int	Required, value between 1 and 100

Student

Property	Data Type	Validation Rules
Id	int	Primary key
FirstName	string	Required, maximum length 50
LastName	string	Required, maximum length 50
Email	string	Required, valid email address, maximum length 150
PhoneNumber	string	Optional (nullable), maximum length 20, valid phone number if provided
DateOfBirth	Date	Required

Enrollment

Property	Data Type	Validation Rules
Id	int	Primary key
StudentId	int	Required foreign key
SubjectId	int	Required foreign key
EnrollmentDate	DateTime	Required
Grade	decimal	Value between 0 and 100

Relationships

Relationship	Type	Description
Department → Teacher	One-to-Many	A department can have many teachers. Each teacher belongs to one department.
Teacher → Subject	One-to-Many	A teacher can teach many subjects. Each subject is assigned to one teacher.
ClassRoom → Student	One-to-Many	A classroom can contain many students. Each student belongs to one classroom.
Student → Enrollment	One-to-Many	A student can have many enrollment records.
Subject → Enrollment	One-to-Many	A subject can have many enrollment records.
Student ↔ Subject	Many-to-Many using Enrollment	A student can study many subjects, and a subject can have many students.

Data Annotations

- Apply Data Annotations to enforce basic validation rules on entity properties.
- Use Required validation where specified.
- Apply maximum string lengths according to the model specifications.
- Apply EmailAddress validation to email properties.
- Apply Phone validation where a phone number is provided.
- Apply Range validation to numeric properties where specified.
- Use nullable properties for optional fields.